

The Role of Futures Market in Aggravating Commodity Price Inflation and the Future of Commodity Futures in India

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Abstract

Persistent inflationary pressures in global commodity prices in the recent past sparked a debate over its nature with speculation in commodity markets being singled out as the primary factor behind rising prices, even leading to a demand for a ban on futures trading for several important commodities. In recent times, increased amounts of capital have been flowing into the commodity futures trade, and there is thus a need to analyse the role futures market participants can possibly play in forming or distorting prices in the market for the underlying commodity. Investigations carried out by the US Commodity Futures Trading Commission and the Indian Expert Committee on Futures Trading could establish no conclusive proof regarding the role of the futures market in aggravating inflationary pressures. However, the task forces have again brought forward some important issues, which can help form a guideline for improving infrastructure, surveillance and efficiency in the commodity futures markets in India.

I. Was Futures Trading Stoking Inflation?

Commodity prices rose sharply since the summer of 2007, across the energy, agricultural and metals complex. The persistent inflationary pressures in global commodity prices sparked a debate over its nature. On one hand, speculation in commodity markets is being singled out as the primary factor behind rising prices. On the other hand, many observers have opposed this point of view, arguing that there is no evidence, either empirical or theoretical, that speculative activity is the sole reason behind this present bout of inflation. The debate over the role of the futures market in causing the recent global inflationary trend is primarily based on the premise that the failure of financial markets since the sub-prime crisis last summer has led to increased speculative activity in the commodity sphere, which has remained rather profitable. While it is true that in recent times increased amounts of capital have been flowing into commodity futures trade and more so due to both the need to cover losses in financial markets and to hedge against future inflation, there is a need to not only analyse the gamut of supply and demand side factors leading to sustained high levels of inflation, but also to understand the role participants can possibly play in the

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maintenance or via
financial
instruments such as
futures or options
contracts.

market, rather than associate higher levels of futures activity directly with mispricing in the futures markets. The need to probe further is strengthened in the Indian context by the ineffectiveness of the ban on wheat futures in containing rising food (wheat) prices.

In this article we try to analyse the role of the futures market in aggravating commodity price inflation. For this purpose we first take a look at the role of the futures market and the different participatory entities in the commodity futures market to see what possible contribution they can make in price formation (/distortion). Next, we look into the findings of a study by the US Commodity Futures Trading Commission (CFTC) and the Indian report submitted by the Expert Committee on Futures Trading (ECFT) in response to a demand for a ban on futures trading for several important commodities. We find that no conclusive proof could be established regarding the role of the futures market in this context, either in India or in the US, where similar accusations have been made. However, these task forces and some other studies on commodity futures markets have brought forward some important issues, which can help form a guideline for improving infrastructure and surveillance in the commodity futures markets, particularly in a developing market like India.

II. Some Dynamics of the Market

How the Market Functions

Central to the theme of a commodity futures market is the dynamics of commodity prices, production, and inventories, as well as the sources and effects of market volatility. Commodity markets tend to experience varying levels of price fluctuations as well as volatility at different time periods due to changing supply, demand and inventory conditions. Price volatility drives the demand for hedging, which could be done through physical storage (inventory maintenance) or via financial instruments such as futures or options contracts. While physical storage requires infrastructure and may be quite costly from an individual point of view, operating in the futures market provides a less costly and more efficient way to reduce future price risk through hedging and trading of risk.

Commodity futures are part of the derivatives family of financial products as their value derives from the underlying instrument. Futures contracts are financial instruments that carry with them legally binding obligations and are traded on regulated futures exchanges. These contracts are standardised in terms of quality, quantity and settlement dates. Futures contracts are traded on organised exchanges and the futures contract is *marked to market*, which means that there is a settlement and corresponding transfer of funds at the end of each trading day. Buyers and sellers have the obligation to take or make delivery of the underlying commodity at a specified price on the settlement date in the future. Commodity derivatives exchanges provide a platform where traders and investors from various parts of the country

(/world) can participate in the hedging and price discovery of any listed commodity. We enlist below some features of commodity derivatives markets, which shed light on the functioning of the market and also help to understand its role in price formation of the underlying commodity.

Some Features of Commodity Futures Trading:

- Exchanges are regulated by a government authority e.g., Financial Services Authority (FSA) in the UK, the Securities and Exchange Commission (SEC) in the USA, and the Forward Market Commission (FMC) in India.
- Each futures exchange has a clearing house, which ensures that trades are settled in accordance with market rules and that guarantees the performance of the contracts traded.
- To trade on an exchange, one needs to be a member of that exchange. Exchange members can trade on their own account or acting as brokers they can execute orders for other investors.
- In an exchange, buyers and sellers of a contract express their demand and supply; trading or price matching can take place through electronic dealing systems, open outcry around a pit or a combination of both.
- When market participants buy futures, they do not pay the full amount of value of the contracts they purchase. Rather, they pay an initial margin that acts like an insurance deposit (the amount is determined by the clearing house). This initial margin represents a percentage of the value of the transaction.
- At the end of each trading day, individual positions are evaluated relative to the closing price of the market published by the exchange; participants are then said to be marked to market. If their position is profitable, that profit will accrue into their account. In contrast, if the position is not profitable, the loss will be deducted from the initial deposit and the participant will be given a margin call (called the variation or maintenance margin) to make up the difference.
- On the settlement date or the expiry of a futures contract, the buyer and seller have the obligation to make or take delivery of the instrument. Settlement can be carried out in two ways: through the actual delivery of the commodity into a predefined location, or through a cash settlement, whereby the value of the position is assessed relative to the settlement price and a corresponding financial payment is made.
- In reality, very rarely does physical delivery take place in commodity futures. At the same time, market participants do not necessarily need to wait for the expiry of their contract to settle their obligation vis-à-vis the exchange. Positions are often closed by taking an offsetting position for an equal and opposite amount of contracts.

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Thus commodity futures trading takes place in a well-regulated environment and even though commodity futures transactions are leveraged, daily margin requirements ensure that individual member positions are monitored and evaluated on a regular basis. A point to note from the above is that operations in the commodity futures market need not directly involve transactions in the commodity in question. Although futures and forward contracts specify prices to be paid at the time of delivery, it is not necessary to actually give/take delivery of the commodity. In fact, the vast majority of futures contracts are *closed out* or *rolled over* before the delivery date; so the commodity need not change hands at all. This is true irrespective of whether the contracts are held for hedging or speculation purposes. The requirement for efficient arbitrage between physical and financial markets is provided by the obligation to either deliver or receive the physical commodity if a contract is held to expiry. However, there is no obligation to hold a contract to expiry and in fact very few users of commodity exchanges ever take physical delivery. Most are involved in risk management or financial transactions and either close out or roll their positions forward before they expire.¹

As in the case of all financial markets, commodity futures price is also a function of demand, supply and market sentiments. Commodity futures prices reflect the price that both the buyer and the seller agree will be the price at the time of delivery. Therefore, these prices provide direct information about investor's expectations about the future price of the commodity. Like the prices of every other risky asset, futures prices also include risk premiums, to reflect the possibility that spot prices at the time of delivery may be higher or lower than the contracted price. In the absence of any manipulative activity, the futures markets provide estimates of the demand/supply situation of a particular commodity in the near future. Commodity futures prices go up when market participants think that the supply of the commodity would not be able to meet the demand in future. On the other hand, if the expectations for a surplus production or reduced demand exist in

¹ For example, an industrial consumer of crude oil, worried about the risk of oil price increases during the coming year, may take a long futures position in crude oil in January, by buying, say, an appropriate number of July futures contracts, but may continue to buy oil from his usual source. If the price of oil rises between January and July, the consumer will pay more for his oil, but will enjoy an offsetting gain from the futures position. Likewise, if the price goes down, the consumer will pay less for oil but have an offsetting loss from the futures position. As July approaches, the consumer might roll over his position by selling the July contracts and buying, say, December contracts. As December approaches, the consumer may roll over the position again, or simply close it out by selling the contracts. Throughout, the consumer buys oil in the spot market and never takes delivery on the futures contracts. Likewise, an oil producer concerned about the risk of oil price decreases could hedge this risk by taking a short position in oil futures. Any decreases in oil prices would then be offset by gains from the futures position.

the market, future prices of the commodity tend to decline. This information is very important for policymakers as it enables them to take appropriate action so that the demand-supply gap can be filled in time.

The Crucial Link with the Physical Market

Commodities are different from financial assets as they involve storage; it is possible to derive a fundamental relationship between the physical commodity market and the market for storage through the futures market. In markets for storable commodities, inventories play a crucial role in price formation of the commodity. Futures prices provide important information about spot and storage markets. Since most commodities exhibit inelastic supply in the short run, holding physical stocks often provides some advantages or flexibilities for manufacturers in managing their operational risks. The benefits derived by a holder of commodity inventory (net of storage costs) is termed *convenience yield*, which is reflected as a premium (mostly positive) in the spot price.² *Cost of carry* for the commodity is the total cost of storing, namely the physical storage cost plus an opportunity cost in terms of the forgone interest.

The relationship between the spot and futures price indicates the need for storage and production. The futures price will be greater or less than the spot price, depending on the magnitude of the net storage costs (marginal convenience yield). If the marginal convenience yield is large, the spot price will exceed the futures price; in this case the futures market is said to exhibit strong *backwardation*. If the marginal convenience yield is precisely zero, the spot price will equal the discounted future price. If the net marginal convenience yield is positive but not large, the spot price may be less than the futures price, but greater than the discounted future price. The price spread between the spot and futures market, termed the *basis*, thus helps determine the levels of storage; as the basis moves above the net carrying cost, it becomes profitable to buy in the spot market and sell in the future.³ Convenience yield rises when volatility increases because greater volatility increases the demand for physical storage as commodity users will need greater inventories to buffer fluctuations in production and consumption. As basis volatility (risk) increases the effectiveness of a futures hedge decreases.

² The benefits arise from the use of inventories to reduce production and marketing costs, and to avoid sudden shortages. Production as well as inventory buildup (/drawdown) decisions are made in the light of two prices: a spot price for sale of the commodity itself, and a price for storage. This price of storage is equal to the marginal value of storage, i.e., the flow of benefits to inventory holders from a marginal unit of inventory, and is termed the marginal convenience yield.

³ See Pindyck (2001) for a detailed exposition.

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Role of Hedgers, Traders and Speculators

Several participants operate in the futures market, where buyers and sellers of contracts express their demand and supply for the contract and prices are matched to arrive at a fair market price. Hedgers come to the market with a view to offsetting the price risk inherent in any cash market position by taking an equal but opposite position in the futures market.⁴ On the other side, there are the traders or speculators/arbitrageurs, including banks and other financial institutions, with a view on the direction commodity prices will take. They assume the risk and provide liquidity to the market. The traders or arbitrageurs prefer an immediate view of the market and these divergent views lead to price discovery for the commodity concerned. Contrary to popular belief, successful speculators actually promote price stability in markets. By buying low and selling high, speculators push up the low prices and push down the high prices and the presence of speculators thus reduces price volatility.

The opinions about the ill-effects of speculation in commodity on prices are based on the argument that a large and sudden increase in the market position of any subset of market participants will tend to move prices up if the increase is in demand (expressed in financial markets as an increase in long positions) and down if the increase is in supply (expressed as an increase in short positions). Even though there is a long for every short and a short for every long, as contracts need to be matched when a price is arrived at, large open interest positions can push up futures prices and thus influence spot prices.⁵ Globally, and particularly in the US oil market, activities of hedge funds, investment banks and index funds have been the subject of scrutiny.⁶ In the recent debate on the oil market, the bubble argument suggests that developments in financial energy markets (especially the increase in cash under

⁴ This technique is very useful in the case of any long-term requirements for which the prices have to be firmed so as to quote a sale/purchase price, but the hedger wants to avoid buying the physical commodity immediately to prevent blocking of funds and incurring large holding costs.

⁵ For example, if on a given day an extremely large speculator decides to go short, his brokers will then attempt to purchase a large number of short contracts. The order for say, 5,000 short contracts amounts to a search for 5,000 long contracts, and in the open outcry/order matching process the bid price may fall until the necessary number of longs is attracted to take the offsetting positions for the 5,000 shorts. All other things being equal, the effect of such a large increase in the number of shorts demanded is to drive down the price. If the intra-day price decline forces prices below a technical support level, those who trade on such signals will be attracted into the market, creating further downward price pressure.

⁶ Apart from hedge funds, commercial and investment banks make a variety of offerings to investors that, ultimately, result in a financial institution placing substantial hedge positions in the market. For example, some banks offer structured notes indexed on oil with fixed or guaranteed returns. Not all buyers of these notes are oil market participants, and may purchase them largely for portfolio diversification. Of greatest recent interest is the role of index or passive investors, who are looking for portfolio insurance via commodity returns, and are prepared to enter the market at any price level.

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management of hedge and other funds, and the decisions of index-oriented funds to take long positions in commodities, including energy) may have precipitated a classic condition of too many buyers chasing too few sellers of financial oil (/commodity) instruments.⁷ In India, the price rise in commodities has been attributed to the role of the futures market even though large institutional operators are not yet allowed in the market. It has been alleged that accumulated net long positions, in effect constituting a bet that prices would rise, actually affected prices.

There is no economic justification in clubbing all speculative activities and no reason why they should all work in one direction; even if large funds are operating in the market different fund managers take positions according to their own portfolio management needs and have differentiated trading strategies, which in fact adds to market liquidity.⁸ However, similar moves by a large group of participants are possible under certain circumstances. The most obvious situation is when the fundamental supply and demand situation clearly points in

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⁷ The overall assets under management (AUM) of commodity indices have risen from negligible size in 2003 to an estimated \$76.7 billion in January 2006 and ballooned to \$297 billion by June 2008, according to an estimate by Lehman Brothers. However, of this \$219.3 billion increase since 2006, \$98.1 billion has been new financial inflow, with the remainder due to the appreciation of the underlying commodities. It has been pointed out (Cooper *et al*, 2008) that a bulk of the increase in commodity index AUM has been due to the surge in commodity prices. As the indices rise, so does the value of index linked assets. That is an automatic accounting change that does not represent any new inflow of money, or indeed any intervention or activity at all by investors. The value of commodity index AUM is estimated to have risen by \$13 billion in Q1 2008, and of this \$11 billion was simply due to the increase in the underlying indices. The net inflow of new money over the quarter was then about \$2 billion.

⁸ There are thousands of trading strategies followed by each group of participants. These are usually based on the following approaches: *Technical approach*: here traders use technical analysis, which means that they monitor chart patterns and expect that they will repeat themselves in the future. The most common technical indicators usually include moving averages and strategies developed on the idea of a break through a specific price barrier. *Fundamental approach*: this strategy is usually adopted in addition to some technical analysis. Traders do not directly follow historical data or price ranges; instead they react upon the fundamental situation on the market. They trade upon news, inventory levels, weather forecasts (which affect the underlying commodity), i.e. they focus on the supply and demand situation. *Systematic approach*: advanced mathematical and statistical methods are used in order to develop fully automated programs, which then generate trading signals and eventually transmit orders for execution. *Trend following*: this strategy has the longest history and continues to be implemented most often. Positions are opened in the current market direction. Naturally, these traders need trending markets, i.e. those with long and uninterrupted trends without major corrections. *Momentum trading*: momentum is a general term used to describe the speed at which prices move over a given time period. Momentum indicators determine the latent strength or weakness of a trend as it progresses over time. Momentum is highest at the beginning of a trend and lowest at trend turning points. Any divergence of directions in price and momentum is a warning of weakness; if price extremes occur with weak momentum, it signals an end of price movement in that direction. If momentum is strong and prices are flat, it signals a potential change in price direction. *Countertrend strategy*: traders who adopt this less-common strategy try to anticipate the price top/bottom and initiate a position against the current trend.

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one particular direction. On the other hand, such effects can be expected if a large market participant is exercising market power, or subsets of market participants are (deliberately or coincidentally) acting together. In financial markets, investors have access to common information sets and may employ similar techniques in evaluating this information. Therefore, agents may behave in a similar manner, not through consciously following the actions of others, but through acting upon the same information. However, such behaviour, if deliberate, comes under the category of market manipulation and would lead to inefficiency in the market. Herding is a behavioural pattern in which there is a deliberate attempt by agents to mimic the actions of others.⁹ The type of herding most directly related to the context of futures market traders is based on the theory of information cascades (Banerjee, 1992). An information cascade arises when decisions are made by each agent sequentially, but agents begin to ignore their private signals in favour of the observed actions of previous agents. The sheer weight of numbers may cause agents to discard their private information and use the decisions of others to herd.¹⁰ Herding equilibrium may not be socially efficient and prices may be more volatile than if agents had acted independently of each other, as individual information and decision making is forgone in following the herd. The possibility of herding can lead to multiple equilibria, causing asset prices to deviate for prolonged periods from fundamentals.¹¹

⁹ Herding may arise for a variety of reasons. Managerial remuneration often depends upon achievements and reputation; thus, poor managers have an incentive to copy the decisions of other managers in order to mask their inferiority. Agents may also be compensated according to performance relative to their peers; thus risk-averse managers will be unlikely to deviate from their peers, and will tend to cluster around their peers' portfolio decisions.

¹⁰ In commodity markets, the inelasticity of supply and demand in the short run causes sharp price responses to small physical imbalances, making it difficult to formulate price expectations. Opacity of information or informational vacuums regarding, say, actual oil supplies etc. also make decision making difficult and promote herd behaviour. While the theoretical literature on herding is well developed (see Bikhchandani and Sharma, 2001, for a review), the empirical literature has performed only indirect tests of the various herding theories. A major limitation in any test of herding is the inability to separate intentional herding from coincidental decision making, in which agents may appear to make similar decisions through possessing similar information, resulting in correlated decisions, but this would not imply the existence of herding. In an important study, Adrangi and Chatrath (2008) test for herding using SEC (COT) data on aggregate trader positions for four commodities over twenty years (up to 2002). They show that while the positions of commodity traders are highly related, the relatedness falls short of herding. The cross-commodity relatedness in trader positions is almost entirely explained by common demand and supply factors.

¹¹ Banerjee (1992) notes that the sequential decision process implied by herding, reduces information in the market. At the very least, if the herders' information is flawed, herd behaviour can lead to short-run mispricing, impeding the decision-making abilities of hedgers. Anonymity of markets allows participants to potentially misinterpret an uninformed investment as a bullish move by a trader with superior information. If participants then herd around this investment, prices can shift to a new non-equilibrium level, till other stronger signals are received.

However, irrespective of the dynamics of the futures market, the underlying commodity prices driven by manipulation or misconception are unlikely to sustain for too long if they did not match the fundamental demand and supply situation. As has been argued (Murphy, 2008; Krugman, 2008), if speculative investors push up the futures price of a commodity, say oil, which in turn will drive up the spot price of oil, higher prices would in the medium to long run drive up supplies of physical oil. If commercial demand is lower, there would be an excess supply because the higher world price of oil would have encouraged producers and discouraged consumers of oil. If speculators really have driven up the world price of oil above the level justified by the fundamentals, then world output should be exceeding world consumption leading to physical hoarding in the longer run. On the other hand, if there is no hoarding, meaning that commercial consumers are purchasing all the output, then the higher price is justified by the fundamentals. In effect, if speculative activity drives market prices above the level at which supply and conventional demand are matched, then the data should indicate a growing stockpile of excess supply (or restricted output). For this kind of situation to occur, the markets have to be in contango or futures price above spot and sufficiently so to make storage worthwhile. This was clearly evident in the data on the housing boom, but did not appear at all in the data on oil inventories.¹²

Apart from traditional hedgers and speculators, the role of passive investors such as index funds has particularly been mentioned as a prime cause behind the present bout of price pressures felt in the US. The reason cited is that index speculator demand is distinctly different from traditional speculator demand as it arises purely from portfolio allocation decisions.¹³ The allegation is that, when an Institutional Investor decides to allocate, say, 5 per cent to commodity futures, they come to the market with a set amount of money. They are not concerned with the price per unit; they will buy as many futures contracts as they need, at whatever price is necessary, until all of their money has been put to work. It was alleged their insensitivity to price multiplies their impact on commodity markets. However, on the contrary, even for passive investors the needs of portfolio management require that a fund adjust its allocation on the commodity futures as and when price changes occur (Murphy, 2008). For example, if a fund has decided on, say, a 5 per cent allocation in oil futures, according to its portfolio needs, when the price of oil rises, the fund's total value

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¹² During the housing boom, there was a larger-than-normal fraction of rental units that were vacant, because investors were pushing prices higher than the fundamentals, in hopes of gaining from future appreciation. Whereas, the oil market data showed that during the period in question, inventories were not growing, and the futures market was characterised either by backwardation—futures price below spot—or by a contango too weak to make storage profitable.

¹³ These investors distribute their investment across the key commodities futures according to the popular indices: the Standard & Poors-Goldman Sachs Commodity Index, and the Dow Jones-AIG Commodity Index.

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increases, but the share of oil futures rises disproportionately, so that the allocation is now higher than the desired 5 per cent. If the price hike has not changed the fund's underlying views about the future of the market, then during the next rebalancing the fund will reduce its holdings of commodity futures, i.e. it will become a net seller. Again, the opposite holds if commodity prices fall; then the fund will buy additional futures contracts in order to restore its desired 5 per cent allocation. The benefit of this process for market functioning is that it provides additional liquidity to the futures markets. The investor can however, increase his allocation to commodities from, say, 5 per cent to 8 per cent, but that decision would be made on the basis of his expectations about the trends in the commodity futures market.

III. The US & Indian Investigations into the Role of the Futures Market in Aggravating Commodity Price Inflation

In this section, we outline the mode of investigation and the findings regarding the role of the futures market in the persistence of high inflationary pressures in the US and in India. The US findings are important as the US being an important mature market has substantial influence on price formation in the global and hence Indian futures markets. The US investigations also provide some guidance on structural, regulatory and surveillance issues that need attention in a nascent market like India.

US

The US Commodity Futures Trading Commission (CFTC) was urged to investigate the happenings in the futures market as grievances were raised by both consumers and agriculturists. The CFTC opened an investigation into possible price manipulation and abuse in agricultural commodities and crude oil futures, as futures prices for crops like corn, wheat, soybeans and cotton surged to record levels since 2005.¹⁴ It was alleged that inflation was driven by a demand shock coming from a new category of participants in the commodities futures markets, namely, Institutional Investors, as these Corporate and Government Pension Funds, Sovereign Wealth Funds, University Endowments and other Institutional Investors collectively account on average for a larger share of outstanding commodities futures contracts than any other market participant (CFTC, 2008a). It was also pointed out that the CFTC has invited increased speculation by granting Wall Street banks an exemption from speculative position limits when these banks hedge over-the-counter swaps transactions.¹⁵ This has effectively opened a

¹⁴ See *Appendix A* for a summary of the various demand and supply side influences working on commodity prices during the aforementioned period.

¹⁵ When Congress passed the Commodity Exchange Act in 1936, they did so with the understanding that speculators should not be allowed to dominate the commodities futures markets, but CFTC has allowed certain speculators virtually unlimited access to the commodities futures markets.

loophole for unlimited speculation, as, when Index Speculators enter into commodity index swaps, which 85-90 per cent of them do, they face no speculative position limits.

The US inquiry was intended to review the influence of speculative investors on energy and food prices and particularly the role of hedge funds in fuelling inflation. The questions asked were:

Do supply and demand factors justify current prices, and if not, what other factors may be in play? Do contract terms and conditions need to be revisited by exchanges? What is the impact of financial players such as speculators and index traders on the functioning of the markets and the commercial players?

To carry out the investigations, the CFTC used its data on Large Trader reporting System, under which clearing members, futures commission merchants (FCMs), foreign brokers, and individual traders file confidential reports with the CFTC every day, reporting positions.¹⁶ When a reportable trader is identified to the CFTC, the trader is classified either as a *commercial* or *non-commercial* trader. A trader's reported futures position is determined to be commercial if the trader uses futures contracts for the purposes of hedging as defined by CFTC regulations. Traders who are non-commercial include hedge funds, commodity trading advisors, commodity pool operators (managed money traders), and floor brokers and traders.

The main findings of the investigations are (CFTC, 2008a&b; ITF, 2008):

There is little economic evidence to demonstrate that prices are being systematically driven by speculators in either oil or agricultural commodity markets. Generally, the data shows that:

- Prices have risen sharply for many commodities that have neither developed futures markets (e.g. Durham wheat, steel, iron ore, coal, etc.) nor institutional fund investments (Minneapolis wheat and Chicago rice).
- Markets where index trading is greatest as a percentage of total open interest (live cattle and hog futures) have actually suffered from falling prices during the past year.
- The level of speculation in the agriculture commodity and the crude oil markets has remained relatively constant in percentage terms as prices have risen.

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¹⁶ CFTC identifies each large trader in each designated contract market (DCM), for example, in the NYMEX WTI crude oil futures contract a trader with a position exceeding 350 contracts in any single expiration is reportable. Large-trader positions reported to the CFTC consistently represent more than 90 per cent of total open interest in the NYMEX WTI contract, with the remaining traders carrying smaller positions.

There is little evidence that daily position changes by any of the trader sub-categories systematically precede price changes. This result holds for all potential categories of speculators.

- Studies in agriculture and crude oil markets have found that speculators tend to follow trends in prices rather than set them.
- Speculators such as managed money traders are both buyers and sellers in these markets. For example, data shows that there are almost as many bearish funds as bullish funds in wheat and crude oil.

However, it has been pointed out that the *swaps loophole* allows all speculators to use it to access the futures markets; in the CFTC's classification scheme, all speculators accessing the futures markets through the swaps loophole are categorised as commercial rather than non-commercial.¹⁷ The CFTC requested additional reporting and then classified positions by entity (commercial versus non-commercial) and not by trading activity (speculation versus hedging).¹⁸ In the crude oil futures market, the evidence again suggested that changes in speculative positions follow the reactions of commercial traders to relevant news, so that commercial rather than speculative position changes are driving price changes. If a group of market participants has systematically driven prices, detailed daily position data should show that that group's position changes preceded price changes. The analysis of daily price changes and position changes by various trader groups and combinations of trader groups between January 2003 and June 2008 shows that over the time period, there is little evidence that daily position changes by any of the trader sub-categories systematically precede price changes. This result holds for all potential categories of speculators—for non-commercial traders in total, for hedge funds and swap dealers individually, and for the positions of non-commercial traders combined with swap dealers (*Exhibit 1*). It was also found that the number of non-commercial traders exceeding stipulated limits was only 6 and among them some held long positions while some held short positions, negating the allegation of build-up of extensive long positions in the oil futures market.

The CFTC maintained that much of the increased inflow is speculative, in the sense that it is in anticipation of future supply constraints and robust demand. Both have been very much in evidence in recent years, and to the extent that speculation is driven by such factors it is playing a proper and indeed important role; that is, signalling the need to expand investment in production capacity, and providing liquidity to hedgers. There is very little evidence that this inflow is manipulative. Low and declining levels of inventory for major food

¹⁷ If a hedge fund wants a \$500 million position in wheat, which is way beyond position limits, they can enter into swap with a Wall Street bank and then the bank buys \$500 million worth of wheat futures.

¹⁸ These trader classifications have grown less precise over time, as both groups may be engaging in hedging and speculative activity.

EXHIBIT 1

Granger Causality Tests Relating Daily Position Changes to Price Changes in the NYMEX WTI Crude Oil Futures Contract from January 2000 to June 2008

Trader Classification	Direction of Causality	
	Price Changes lead Position Changes	Position Changes lead Price Changes
All Commercials (includes Manufacturers, Commercial Dealers, Producers, Other Commercial Traders, and Swap Dealers)*	+ve Significant (p-Value 0.028)	Not Significant (p-value 0.896)
All Non-Commercials (includes Hedge Funds, Floor Brokers & Traders)	Not Significant (p-value 0.062)	Not Significant (p-value 0.764)
Hedge Funds	+ve Significant (p-Value 0.003)	Not Significant (p-value 0.585)
All Non-Commercials combined with Swap Dealers	Not Significant (p-value 0.062)	Not Significant (p-value 0.947)
* Each of the categories tested individually gave the same results of positive causality in the first direction and non-significant causality in the second. <i>Source: ITF (2008).</i>		

crops, for example, indicate no potentially manipulative hoarding going on in that sector. There is also no clear relationship between the small changes in numbers of futures equivalents related to commodity index business and the movement or volatility of wheat futures prices.¹⁹ Importantly, as a result of this survey, the CFTC recommends improvements to the classification process and reporting requirements for large traders that will help the agency better quantify the nature and accuracy of the trading activity being conducted on exchanges.

India

In the Indian market on the contrary large institutional players have not been allowed to operate in the commodity futures market. However, under the suspicions that manipulative activity was causing distortions in the futures market and stoking inflation, the Indian Government bestowed the responsibility of a study on the effects of futures trading on inflation in the country on the Expert Committee on Futures Trading (ECFT). The Indian investigations were mainly based on comparison of commodity price trends and volatility pre- and

The Indian investigations were mainly based on comparison of commodity price trends and volatility pre- and post-futures and also took into consideration the production and supply of the crucial commodities in the relevant years.

¹⁹ Both the net notional values and the equivalent numbers of futures contracts reported for commodity index trading in wheat changed very little over that time period through which Wheat futures prices experienced a great deal of volatility. The nearby futures price was at around \$8.85 per bushel on December 31, and traded near \$13.00 in late February and early March, before declining to \$8.44 at the end of June, while the index notional value increased by about 7 per cent.

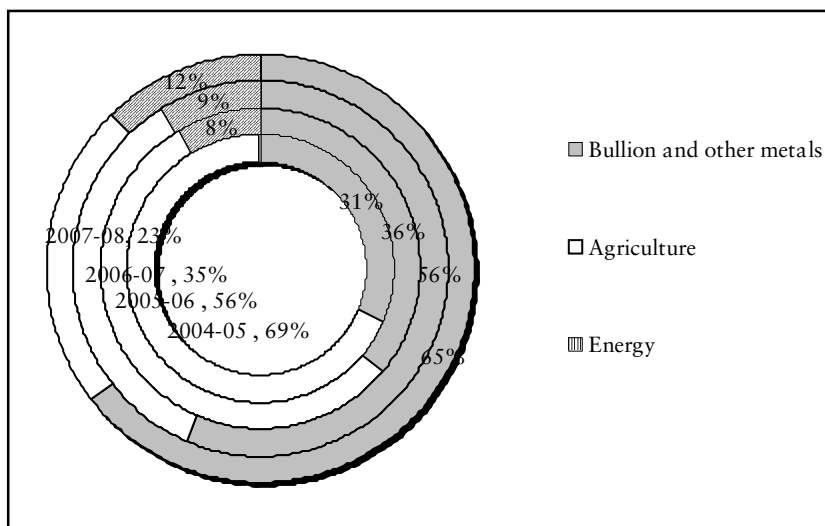
post-futures and also took into consideration the production and supply of the crucial commodities in the relevant years. The terms of reference of the ECFT were as follows:

- (i) To study the extent of impact, if any, of futures trading on wholesale and retail prices of agricultural commodities; and
- (ii) Depending on (i), to suggest ways to minimise such an impact;
- (iii) Make such other recommendations as the Committee may consider appropriate regarding increased association of farmers in the futures market/trading so that farmers are able to get the benefit of price discovery through Commodity Exchanges.

Unlike the CFTC report, the Indian study is primarily a study on price trends in commodities, in which futures have been allowed in comparison to other commodity groups. The main findings of the ECFT were:

In terms of volumes of trade, although agricultural commodities led the initial spurt and constituted the largest proportion of the total value of trade till 2005-06 (55.32 per cent), this place was taken over by bullion and metals in 2006-07 (*Chart 1*). Further, there has been a fall in agri-commodity volumes during 2007-08 over the previous year (Para 3.2 in Report).

CHART 1
Share of Commodity Groups in Trade



As for foodgrains, food items and agri-commodities, a general rise in the prices of these groups of commodities was observed, which was higher than that of the entire WPI or CPI. As the weight of food items, particularly of foodgrains, is much higher in the consumer price indices as compared with the WPI, the contribution of these commodities to CPI inflation is correspondingly higher.

EXHIBIT 2 Contribution of Agricultural Commodities to WPI &CPI Inflation (%)				
Price Index	Overall Rise in Index	Rise in Food Index	Weight of Food in Index	Contribution of Food to Inflation
WPI	6.37 (Jan-07)	10.85 (Foodgrains)	5.01	8.34
WPI	6.37		25.65 (87 Agri- commodities)	31.54
CPI-UNME	7.8 (Feb-07)	11.85	45.61	67.0
CPI-IW	7.6	12.2	57.0	74.0
CPI-AL	9.8	11.8	69.15	83.4
Note: Table compiled from ECFT Report.				

Thus in early 2007, both *food* and *all agricultural commodities* show higher inflation than overall WPI inflation (*Exhibit 2*). The ECFT opined that although this supports the view that the inflation in early 2007 was led by agricultural commodities, it is not possible to conclude that factors particular to these commodities were the only, or even major, reason behind the spurt in inflation, as manufactured products with a weight of 63.75 per cent in WPI also recorded inflation of around 6 per cent.

In order to examine whether futures trade could have led to price rise in agricultural commodities, ECFT has relied on WPI data as these are a closer proxy of producer prices of agricultural produce than retail prices. A total of 21 commodities, with weight of nearly 70 per cent in agricultural futures trade, have a weight of only 11.73 per cent in the total WPI basket and account for less than half of the weight of the 87 processed and unprocessed agricultural commodities that are included in the WPI (Para 4.9). Both monthly and weekly data show that the annual trend growth rate in prices was higher in the post-futures period in 14 of these commodities, (viz. chana, pepper, jeera, urad, chillies, wheat, sugar, tur, raw cotton, rubber, cardamom, maize, raw jute and rice); and lower in 7 commodities (viz. soy oil, soy bean, rapeseed/mustard seed, potato, turmeric, castor seed, and gur). However, ECFT points out that a revealing feature of this data is that of the 14 commodities in which price acceleration took place in the post-futures period, 10 had suffered negative inflation during the pre-futures period (*Exhibit 3*). It is possible in such cases that the acceleration in the growth rate of WPI in these commodities is simply rebound and catch-up with the trend, which in turn could have been aided by more efficient price discovery. Similarly, of the 7 commodities in which WPI growth was lower post-futures, 6 had unusually high pre-futures inflation at over 10 per cent. In these cases, too, it is possible that what is being observed is simply reversion to a more normal level of inflation. In both cases, there is the problem that the period during which futures markets have been in operation is much too short to discrimi-

Of the 14 commodities in which price acceleration took place in the post-futures period, 10 had suffered negative inflation during the pre-futures period. It is possible in such cases that the acceleration is simply rebound and catch-up with the trend.

Although inflation clearly increased post-futures in some sensitive commodities that have a higher weight in consumer prices indices, it is not possible to make any general claim that inflation accelerated more in commodities with futures trading.

EXHIBIT 3
Pre-/Post-Futures Growth & Volatility in Select Commodities

Commodity	Weights	WPI Trend Growth Rate		WPI Volatility (Monthly)	
		Pre-Futures	Post-Futures	Pre-Futures	Post-Futures
Rice	2.449	-0.4	3.0	3.6	2.5
Wheat	1.384080	2.3	9.6	5.3	7.3
Sugar	3.618830	1.2	3.2	7.7	7.6
Chana/Gram	0.223650	-9.2	20.9	10.6	11.3
Potato	0.256470	28.9	11.7	49.6	47.5
Soy Oil	0.178380	21.8	-1.6	14.1	6.1
21 Commodities	11.730770	4.15	5.05	3.95	3.57
Primary Agricultural Products	21.54	4.19	4.99	3.64	4.49

From Table 2B, ECFT.

nate adequately between the effect of opening up futures markets and what might simply be normal cyclical adjustments. Although inflation clearly increased post-futures in some sensitive commodities that have a higher weight in consumer prices indices, it is not possible to make any general claim that inflation accelerated more in commodities with futures trading (4.11-4.16).²⁰

More definitive is the ECFT's analysis of daily volatilities pre- and post-futures, which shows that daily price volatility in wheat less than halved post-futures; that of soy oil was exactly half; and for urad it had come down by more than 10 percentage points, while it remained same for chana (*Exhibit 4*). The most dramatic result was for a crucial perishable crop like potatoes; the 441 price observations pre-futures showed price volatility of 245.9 per cent, while the same

EXHIBIT 4
Pre- and Post-Futures Daily Price Volatilities for Select Commodities

Commodity	Pre-Futures	Post-Futures	Observations (Pre/Post)
Potatoes	245.9	68.4	441/441
Wheat	43.6	17.0	814/814
Urad	36.5	25.0	312/753
Soybean	7.5	16.0	792/792
Soybean Oil	18.1	9.7	939/939
Chana	22.6	22.6	815/895
Tur	NA	23.5	

Compiled from Table 3, ECFT Report.
Source: NCDEX.

²⁰ In this part of the analysis, the ECFT surprisingly does not use relative prices, but has chosen to compare the absolute prices of commodities in which futures are allowed as against commodities without a futures market.

number of observations post-futures showed volatility of 68.4 per cent.

The ECFT, notably, found that futures prices were in fact indicating price movements in the correct direction. Futures trading in urad and tur, which were quite liquid on the NCDEX platform, were delisted on January 23, 2007. On the date of delisting, four delivery-month contracts, February, March, April and May 2007 were running. The urad futures prices as on January 23, 2007 were in backwardation, predicting a future fall in spot prices. In fact, spot prices did fall after delisting from Rs. 3,551 on January 23, 2007 to Rs. 2,553 on August 4, 2007. As regards tur, except the February 2007 contract, futures prices at the time of delisting were in contango, predicting a rise in spot prices. In fact, spot prices continued their upward trend even after delisting. In the case of rice, futures prices in all contracts were in backwardation at the point of delisting, with the extent of backwardation lower in further contracts, indicating that spot prices were predicted to fall on the arrival of the new harvest in April-May and rise moderately thereafter. The post-delisting spot prices recorded by the NCDEX show that after a brief decline in prices in the post-harvest period of April and May prices started firming up to above Rs. 1,000 per quintal in July and August even though there were no new futures trade in this commodity (4.22-23).

In contrast with the view of critics of futures markets who argue that speculative activity increased with the introduction of such markets and that this in turn led to unusual price movements, the Commodity Exchanges have argued that the price rise that occurred in these commodities before delisting can largely be explained by supply-side factors, involving domestic production and foreign trade. The ECFT also found that price movements are broadly in line with the movements in supply. Real prices of tur were lower during 2005 and 2006 (when supply from production & imports was near normal, while futures trading was significant) than in either 2003 (before futures) or 2007 (after delisting, with below normal supply). This evidence contradicts the claim that futures trading caused excessive increase in tur prices. Since the real WPI of rice declined throughout the period when futures trading was allowed, and increased only after delisting, speculation in futures markets cannot be said to have exerted any strong upward pressure on the spot prices of rice. Urad inflation did flare up very unusually in the period when futures trading was active (August 2004 to January 2007). But this was a period of below normal production, and although higher imports cushioned supply, import unit values rose 48.7 per cent and 37.7 per cent in 2005-06 and 2006-07. Wheat prices did behave unusually and annualised wheat WPI inflation at 9.8 per cent during the 30 months when futures trading was liquid (August 2004 to February 2007) stands in sharp contrast to inflation in either the previous 30 months (1.5 per cent) or in the year subsequent to delisting (0.3 per cent as in February 2008).

Finally, the ECFT report concluded that “the Committee has

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Price increases
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been unable to determine any conclusive causal relationship (between futures trading and inflation) in view of the short time period during which futures markets have functioned and the complexities that arise because a large number of variables impact spot prices”.

Is there any Case for the Ban?

In May 2007, the Indian government, under pressure to control soaring inflation, suspended futures trading in soy oil, potato, rubber and chickpea. Futures trading in rice, wheat and some pulses like urad and tur were already banned in early 2007 for the same reason. The ECFT report clearly does not find any causal evidence between futures trading and commodity price inflation. The report has to rely on information on volumes and prices and volatilities pre- and post-futures, possibly due to the lack of direct data on the type of traders involved and their activity levels and directions, as used by the US CFTC. Price increases have occurred in certain commodities post-introduction of futures, but that could well be attributed to the price discovery role of the futures market as well as supply and demand interactions that could have generally pushed up prices. The futures market has played its role in identifying future demand and predicting prices accordingly, for important crops (like *chana*). The price rises have also been tallied with the supply side figures and barring a few exceptions like wheat, show no abnormal movements (wheat price movements have also been suspect in the US). Liquid futures trading has helped to curtail volatility in prices for certain important agri-commodities like rice and soy oil according to the ECFT figures. Further, futures prices have also given clear directions on the near future spot price swings and thus shown efficiency to that extent, as per the ECFT study.

Apart from the ECFT report we can also look at a few other statistics which show that there is not much rationale behind banning the few commodities that were banned before and after the ECFT report. The risk adjusted returns (for several commodities for which turnovers have been high, making them suspect for malpractices involved in their trade) do not show much reason why abnormal speculative interests may have been generated for these commodities (*Exhibit 5*). There are in fact other commodities like gold which have much higher returns. Again during the period that equity markets gave negative returns, even fixed deposits fared better than many commodities.

The ban has been quite ineffective in consistently softening prices in some commodities like wheat and rubber (*Exhibit 6*). The ban may have in fact deprived Indian consumers of the benefit of global price declines (rice and soy oil) as it has blocked the global price signals. Hence the ban has also underplayed the need to respond to supply constraints or increased demand by adjusting production.

Thus evidence provided by both the Indian (ECFT) and the US (CFTC) inquiries shows that there is no clear case for a ban or curb on

EXHIBIT 5
Returns for Some Commodities Traded on NCDEX

Commodity	<i>Sharpe Ratio (%) (Risk Adjusted Returns)</i>			
	2006	2007	2008 (Jan-Apr)	2005-08 (Cumulative)
Soybean	14.65	31.96	0.95	46.43
Refined Soy oil	37.72	17.23	-3.5	43.02
Chana	8.99	-15.1	1.43	26.27
Guar seed	2.88	-24.64	3.06	3.81
Sugar	-32.19	-43.69	2.08	-41.49
Gold	9.64	8.99	1.98	58.83
<i>Equity</i>	16.14	22.33	-10.18	49.0
<i>Fixed Deposits</i>	6.25	8.25	8.25	28.12
Compiled from NCDEX.				

EXHIBIT 6
Recent Global & Indian Commodity Prices

Commodity	<i>Global prices</i>			<i>Indian prices: WPI</i>		
	2008 over	2008	2008	2008 over	2008	2008
	2007 (April-June)	May over April	June over May	2007 (April-June)	May over April	June over May
Rice	169.9	-0.6	-14.0	8.4	-0.2	0.8
Soy meal	86.1	-0.2	9.7	41.6	6.3	6.0
Soy oil	84.0	0.8	6.0	9.7	3.5	0.7
Wheat, US	68.5	9.2	6.2	8.4	-0.2	0.8
Rubber	31.7	7.5	8.3	36.5	7.0	9.1

futures trading in some essential agri-commodities. In India, neither the volumes of turnover in particular commodities nor their returns are such that one may suspect abnormal levels or directions of speculative activities bordering on market manipulation in them. Post-futures, the movement of prices also does not seem to justify the move. In fact, in the absence of liquid futures markets, prices may give the wrong signals and lead to further misallocation of resources and intensify the supply side constraints that have already been pointed out as a very important cause of the present spate of wide and persistent inflation. However, both sets of reports and several other studies point to some issues, the solutions to which are essential to further the cause of the futures market and ensure better functioning of the market.

IV. The Way Forward for the Indian Commodity Futures Market

Given the fact that the case for a ban on important agri-commodity futures does not stand up to scrutiny, (even the Government must have anticipated the result as it had already included efficiency

In a perfect market with costless delivery at one location and one date, arbitrage should force the futures price at expiration to equal the cash price. Otherwise a violation of the law of one price would exist.

enhancing measures in its terms of reference), one may examine several points on the futures market that have been raised globally during the current high-inflation regime, and take cues to increase the efficiency of the commodity futures market in India.

Issues in Convergence and Efficiency in the Agri-Commodity Futures Market

Convergence between spot and futures prices is an issue because the lack of convergence is often symptomatic of a poorly functioning futures market. In a perfect market with costless delivery at one location and one date, arbitrage should force the futures price at expiration to equal the cash price. Otherwise a violation of the law of one price would exist. In reality, delivery on commodity futures contracts is not costless and is complicated by the existence of grade, location, and timing delivery options.

Real economic damage is associated not so much with increasing basis, but with increased uncertainty in basis behaviour as markets bounce unpredictably between converging and not converging. As first noted by Working (1953), this is damaging because basis in storable commodity futures markets should provide a rational storage signal to commodity inventory holders. A weak basis should be a signal to store and *vice versa*.

Incomplete convergence of futures and cash prices as contract expiration approaches can indicate a poorly functioning market. But an apparent lack of convergence can indicate other issues as well. The US CFTC noted that high storage costs such as can be expected when storage facilities are scarce widen the gap between futures and cash prices. Likewise, a high cost of delivery can create an apparent convergence failure because delivery on a contract incurs both the cost of the commodity and the cost of moving it.²¹ The serious issue of non-convergence and particularly basis volatility, has led the CFTC to look into measures like redesigning of contracts, as well as imposing of limits on speculative positions.

For India, the ECFT noted that a study of the functioning of

²¹ In the US, convergence weakness first surfaced with the July 2006 wheat contract. It was pointed out to the CFTC that non-convergence is extremely large in recent times by historic standards; convergence occurs less often and only for short periods of time. It was alleged that the band, or range, of convergence has widened due to several factors, including: (1) higher and more volatile transportation costs; (2) demand for storage created by biofuels growth; and (3) the futures market running ahead of cash values due to passively managed, long-only investment capital. The basis has become more volatile and weaker than demonstrated historically for corn and soybeans and for wheat more dramatically—thus, convergence has deteriorated. An Illinois University study (Kunda, 2008) showed that results for wheat are different from corn and soybeans, in that basis predictability was unimpressive even before 2006, for wheat. Nonetheless, predictability since 2006 followed the pattern of corn and soybeans and deteriorated substantially relative to the earlier period.

existing futures markets and contracts suggests that although the volume of futures trading in India has increased phenomenally in recent years, its ability to provide instruments of risk management has not grown correspondingly, and has in fact been quite poor. This is so because convergence is not achieved in most cases. The pre-dominance of roll-over type settlements, over delivery-based settlements, is regarded as a cause for such mismatch by many commentators. The ECFT also pointed out that hedging is ineffective in most agri-commodities; this in fact implies a discrepancy between spot and futures prices, saying that futures prices are not able to predict the correct future spot price. (This also means that futures prices even if artificially distorted by speculators have no close link with spot price inflation.) Hedging can reduce price risks of commodity holding if basis risk is less than price risk (i.e. the variance of spot prices), and becomes more attractive the lower the basis risk. A separate study commissioned by ECFT (IIMB, 2008), found that not only was basis risk high for the commodities studied, but this was also higher than price risk for many contracts. Only in the case of one commodity, tur, was basis risk less than price risk in all contracts studied, while in the case of wheat, sugar and urad, basis risk was higher than price risk for a majority of the contracts. Similarly, Lokare (2007) reports basis risk exceeding price risk in a majority of the contracts for gur, potato, rubber, cotton, mustard and wheat; there was no commodity for which all contracts had a lower basis risk than the price risk. This is important since if Indian Commodity Exchanges are offering contracts that are not suited for hedging by holders of physical commodities, not only are these contracts likely to be ineffective in being able to transfer price risk between those holding the commodities and other investors, but the exchanges themselves are also prone to being dominated by undesirable speculative activity. As for the futures market in wheat, Roy (2008) finds an inefficient market mechanism characterised by significant arbitrage opportunity between spot and future markets even at contract expiry. In an earlier study (Bose, 2008), we had found greater convergence for non-agricultural commodities and for commercial products within the agri-commodities, which also showed that there is a difficulty in efficiently predicting spot prices of foodgrains from the futures market.

The ECFT Recommendations

Here we outline the major recommendations of the ECFT (as of many previous commentators) for further development of the Indian commodity futures market:

- Exchanges should act as self-regulatory organisations, capable of demonstrating fair play, objectivity and customer orientation.
- Another enabler of the market will be to upgrade the quality of

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The FMC is not yet bestowed with sufficient powers of market regulations and enforcement. The autonomous status envisaged for the regulator under the Amendment Bill (2007) is designed to provide it with the powers and capacity to intervene in the market more effectively.

regulation both by the Forwards Market Commission (FMC) and by the Exchanges. The proposed FC(R) Amendment Bill to upgrade the regulation and to improve the capabilities of the regulator needs to be pursued vigorously.

- The contract designs should be such that they serve the objective of risk management by farmers and other commercial users.
- Reforming spot markets should also be a top priority. Efficient spot markets would require integration of spot markets, which requires development of rural communication, transport and storage infrastructure.
- For benefits to reach farmers, the support infrastructure of warehousing and commodity finance should be made adequate. This area is likely to undergo a significant improvement after the Warehousing (D&R) Act is operationalised. It is also important that all regulators operating within the commodity market space (like the FMC, Warehouses, Banks and APMCs (Agricultural Produce Marketing Committee)) work in cohesion and do not provide conflicting signals.
- To develop *options in goods*, as they are hedge instruments suitable for farmers' needs.

Regulation and Surveillance

The major lessons to be learnt from the recent investigations are concerning regulation and surveillance issues. Regulation of the US commodity futures market lies with the SEC; through the Securities Act of 1934, the SEC is empowered with broad authority over all aspects of the securities industry. This includes the power to register, regulate, and oversee brokerage firms, transfer agents, and clearing agencies as well as the nation's securities self-regulatory organizations (SROs), such as the stock exchanges. The Act also identifies and prohibits certain types of conduct in the markets and provides the SEC with disciplinary powers over regulated entities and persons associated with them. In India, the regulatory framework for the market is provided in the Forward Contract (Regulation) Act, 1952. The Forwards Market Commission (FMC) set up under this Act regulates the market. Associations organising forward trading have to seek recognition from the FMC. The Rules and Bye-laws of the association are approved by the FMC; however, the FMC, overseen by the Ministry of Consumer Affairs, Food and Public Distribution, is not yet bestowed with sufficient powers of market regulations and enforcement. The autonomous status envisaged for the regulator under the Amendment Bill (2007) is designed to provide it with the powers and capacity to intervene in the market more effectively and enable the commodity market regulator to maintain discipline in the market, on the lines of the securities market regulator, SEBI.

The FMC, on its part, needs to set up a database maintenance system to improve surveillance; the analysis undertaken by the ECFT

underlines the dearth of data on Indian commodity futures market participants. The difference in the methods of analysis in the Indian and US investigations, with the Indian report using available data on prices and supplies and the US CFTC using more direct data on the exact positions of different classes of traders, emphasises the need for a detailed reporting system. The Indian results based on more indirect evidence may suffer from deficiencies and point to the need for greater surveillance in the commodity futures market, which includes maintenance of regular trader based reports. Another lesson from the present US investigations is probably that the regulator needs to be extremely careful about classifying its traders as either hedgers or pure speculators (even though players like index funds are not operational) in order to keep track of their activities and to be able to impose the correct kind of margin requirements and position limits. Particularly, as the US analysis warns that the distinction between pure hedgers and speculators has become quite blurred in present day markets, the need for tracking trader-based data becomes even more important in order to detect market malpractices or malfunctions. Thus the FMC should collect detailed trade data from the exchanges under it and should make public the reports based on its surveillance data, in order to avoid future political pressures for sudden bans and curbs.

Efficiency Enhancement Measures

Quite apart from regulatory worries there are also some other issues that need to be solved in order to make the commodity futures market useful for end-users. Further research needs to be done in the sphere of contract design; this needs multilateral discussions with different types of users of agri-commodities and its futures market. Options in commodities are also an important avenue that needs to be explored as it offers the contract owner greater flexibility, as (s)he has the right but not the obligation to exercise the option.

The need to integrate various local spot markets has been mentioned by the ECFT. Roy (2008) assessed that spot markets for a major produce like wheat in India are cointegrated and LOP (law of one price) rules at the intra-state level market structure, but on a national level, though inter-state markets are cointegrated in the long run, LOP does not hold. The explanation for this can be found in the poor transportation and storage infrastructure, restricted inter-state movement of essential commodities and lack of ICT (Information and Communication Technology) to disseminate the prices across the nation. Similar concerns have been voiced by several commentators. Changes in the Essential Commodities Act (ECA) blocking the movement of agri-produce across the country have been mentioned as a starting point by others. In India, spot/cash market/APMCs/Mandis of commodities are regulated by the respective State Governments/administrative Ministry/Departments, and the FMC only regulates the futures market of the notified commodities through a network of

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national and regional exchanges. The setting up of National Spot Electronic Exchanges by the National Commodity Exchanges is an attempt to create a national integrated market in commodities. The legal and regulatory hurdles in setting up and operationalising these National Spot Exchanges should be looked into. A need to revamp the older and fragmented exchanges had been long emphasised (Thomas, 2003). The older exchanges were seen to be performing price discovery roles even as early as 2001, but there were several regulatory problems that needed to be tackled. Once the spot markets are integrated it would be easier to pin-point the problems that lead to non-convergence between the spot and futures markets and work on them.

As regards inclusion of farmers, it has been noted that world over, farmers do not directly participate in the futures market. They take advantage of the price signals emanating from a futures market. Price-signals given by long-duration new-season futures contract can help farmers to take decision about cropping pattern and the investment intensity of cultivation. Direct participation of farmers in futures market to manage price risk, either as members of an exchange or as non-member clients of some member, can be cumbersome as it involves meeting various membership criteria and payment of daily margins, etc. However, there is no shortage of ideas on how price dissemination can be gradually enabled, through use of telecom and other technology. We must remember that Indian agriculturists have a very long history of dealing in intricate contracts, be it between landless labourers and landowners, or farmers and money lenders.

Warehousing would probably play the single most important role in actually bridging the gap between the farmers and the futures market. Parliament has passed the Warehousing (Development & Regulation) Act, 2007. The Act provides for negotiability of warehouse receipts and this should help farmers to avail themselves of credit lines against their stocks stored in exchange accredited warehouses.²² Commodity exchanges are already into warehousing and assaying, but too much uncertainty regarding major commodity futures trading may hamper the process of the logistics and other facilities being developed. Private partnership would also slow down in the face of such uncertainty. Uniform warehousing facilities and smooth transportation facilities together would ensure that spot and futures prices follow convergence, without which the main role of the futures market of providing price guidance remains unfulfilled.

Theoretical and empirical analyses of financial markets underline the role of high liquidity in fostering efficient markets. Finally, according to the CFTC, profit opportunities result in the buy and sell pressures that bring about convergence, and increased costs of

executing arbitrage trades inhibit arbitrage activities. In the absence of evidence that particular traders may be artificially or intentionally impeding arbitrage in agriculture markets, CFTC cautions against policy choices that might further raise the cost of conducting arbitrage activities.²³ Policies like raising transaction costs and forcing delivery-based trades are very likely to be detrimental to market development in the long run. Liquidity needs to be enhanced such that the actions of a particular interest group do not affect market prices. Liquidity is also essential in reducing basis risk and volatility in the market as less liquid a market, greater the transaction costs and volatility. Hedgers (and arbitrageurs) alone cannot make efficient markets. One example cited is that with the RBI stipulating (in June 2003) that interest futures market can be used by banks only for hedging, the contract never took off. Thus there is a need for a wider range of participants; perhaps the entry of players like banks,²⁴ as envisaged by the FMC, would have more positives than negatives, as long as transparency in operations is maintained.

Appendix A: The Demand- and Supply-side Factors

Here we integrate the results from various studies explaining the factors that could have added to the persistence of inflationary pressures being felt globally. The origins of the boom in oil and food prices have been traced, in detailed studies by the IMF, to the unusually strong global growth in 2003-07. The rapid growth in emerging and developing economies in particular has catalysed demand for commodities, as the industrialisation take-off and strong per capita income increases from a low base are associated with more commodity-intensive economic growth. In the oil market, the strong upward momentum in prices has reflected a sluggish supply response against the backdrop of already stretched spare capacity at the start of the global recovery. The IMF also notes that financial conditions have temporarily added to the upward pressure on the prices of oil and other commodities.

Some financial variables, notably exchange rates, affect prices of oil and other commodities through their impact on physical demand and supply of oil. In contrast, there is little evidence that the increasing investor interest in oil and other commodities as an asset class has affected price trends for oil and other commodities, although purely

Policies like raising transaction costs and forcing delivery-based trades are very likely to be detrimental to market development in the long run. Liquidity needs to be enhanced such that the actions of a particular interest group do not affect market prices.

²³ The CFTC also raised a number of broader issues like the role of suspect economic policies. It noted that Governments subsidise consumption of agricultural staples and energy products, for example, with the effect that demand does not moderate as it should. Governments have also been imposing agricultural export tariffs and bans, with the unintended consequence that farmers are motivated to reduce supply.

²⁴ The possible role of banks has been partly outlined in *Appendix B*.

financial factors, including shifts in market sentiment, may have short-term price effects.

Concerning food commodities, the IMF notes, the recent price surges reflect a confluence of factors. Demand growth, partly reflecting the strong growth in emerging and developing economies noted earlier, has generally outstripped supply growth for many food commodities over the past 8-10 years, notably major grains and edible oils. Global inventories of these crops have thus declined to the low levels last seen in the mid-1970s. The general upward pressure on prices has been strongly reinforced by a number of developments since 2006.

- Unfavourable weather conditions reduced harvest yields in both 2006 and 2007 in an unusually large number of countries. Wheat harvests, in particular, had been adversely affected, which led to a sharp bidding-up of wheat prices, with spillovers into close substitutes (particularly rice).
- Rising biofuel production in advanced economies, in response to higher oil prices, and, increasingly, generous policy support, has boosted demand. In particular, rising corn-based ethanol production accounted for about three-fourths of the increase in global corn consumption in 2006-07. This has pushed up not only corn prices, but also the prices of other food crops, and to a lesser extent, edible oils (through consumption and acreage substitution effects), and poultry and meats (through feedstock costs).
- The rise in oil prices and energy prices more generally has boosted production costs for food commodities, through the impact on transportation fuels and fertilizer prices (the latter have more than tripled since early 2006).
- The growing use of export restrictions by food exporters to raise domestic food supplies and lower domestic prices has put pressure on world prices. Export restrictions by some major rice exporters likely contributed substantially to the run-up in rice prices in 2008.

Investment inflows into commodity assets have surged, as the combination of US dollar depreciation, falling short-term real interest rates, and rising credit risk in advanced economies has made oil and other storable commodities more attractive alternative assets. It is however hard to find concrete evidence that speculation is the main driving force behind the recent increases in commodity prices. For example, one does not see the build-up in inventories that would be the counterpart of rising speculative activity (IMF, 2008).

A similar analysis has been performed by the US CFTC, Task Force on Commodity Markets (ITF, 2008), which also brings out the strengths of the demand side factors that are not driven by purely speculative motives, in influencing oil price inflation.

Appendix B: Role of Banks and Structured Commodity Finance

Standard Credit vs. Structured Commodity Finance

- Standard credit facilities
 - Exposure on balance sheet affects real value of balance sheet
 - Recovery dependent on performance of borrower
 - Difficult to enforce banks' rights
 - Structured commodity finance
 - Externalises the credit risk
 - Commoditises the transaction
 - No exposure to the balance sheet
 - Relevant to new entities without track records
- Thus the latter is a credit risk mitigant for banks.

Utility to Borrowers

- Leverage the strength of commodity by using it as primary collateral
- Structured as off-balance sheet funding
- Structured pricing, which is more attractive than normal working capital or short term loans
- Get financing in tranches aligned with stock build-up schedule
- Repayment schedule aligned with actual usage
- Financing corporates at rates that are linked to commodity prices. E.g., aluminium price linked financing for an aluminium producer
- The structure would help the corporate directly match revenues and interest costs
- Corporate pays lesser interest rates when commodity prices fall and higher interest rates when commodity prices rise.

Warehouse Receipt Based Financing

- Warehouse receipt financing (WRF), through pledge of commodity stocks could help farmers to get a higher amount of bank finance
- WRF can indirectly help farmers through the processing industry as both raw commodities awaiting processing can be financed and processed commodities awaiting sale may be financed
- WRF can facilitate import of raw materials and fertilisers by agricultural/processing units by providing financing facilities.

Restriction on Commodity Exposure of Banks

Banks in India at present are not allowed to deal in commodity derivatives for hedging. Options on commodities are also not allowed.

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2005.

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